Lab 5 Postlab: Metal Evaporation (Flow Cytometer)

Run details. Al was thermally evaporated from a tungsten boat at about 10^{-6} torr in a bell jar about 450 mm across and 600 mm tall. We used SPR220 photoresist instead of the SJR5740 listed in the handout, spun at 2000 rpm for 60 s and soft-baked at about 105 °C for 7 min. At the end of the run, the STC-200 deposition rate controller read THICK = 0.464 kÅ.

1. Aluminum thickness: our estimate vs. the gauge

Gauge reading. The STC-200 showed 0.464 kÅ = 464 Å = **46.4 nm**.

Estimate from the lecture equations. These are the parameters from the TA's Post-Lab 5 update, applied to our chamber:

Parameter	Value	Meaning
m	$(26.98 \times 10^{-3}$ kg/mol) / $(6.022 \times 10^{23}$ atoms/mol) = 4.48×10^{-26} kg	Mass of **one** Al atom, not the total Al loaded
T	1225 °C = 1498.15 K	Evaporation temperature, the middle of the 1200–1250 °C range
P_e	10^{-2} Torr × 133.322 Pa/Torr = 1.333 Pa	Equilibrium **vapor pressure of Al** at T, read from the vapor-pressure chart. This is not the chamber pressure
A	$\pi(1.25\text{ cm})^2 = 4.91 \times 10^{-4}$ m^2	Area of the evaporating source (the same value as the TA's example)
h	0.600 m	Height from the source up to the wafers (our chamber height)
x	0.450 m / 2 = 0.225 m	Sideways offset of the wafer from the source (half our chamber width, as in the TA's example)
ρ	2.70 g/cm^3 = 2700 kg/m^3	Density of Al

Step 1: mass evaporation rate (Hertz–Knudsen).

$$R_{ME} = \sqrt{(m / 2\pi kT)} \cdot P_e$$

$$R_{ME} = \sqrt{[\ 4.48 \times 10^{-26}\text{ kg} / (2\pi \times 1.38 \times 10^{-23}\text{ J/K} \times 1498.15\text{ K})\ ]} \times 1.333\text{ Pa} = \mathbf{7.83 \times 10^{-4}\ kg/(m^2 \cdot s)}$$

Step 2: deposition rate at the wafer. The distance from the source to the wafer is:

$$R = \sqrt{(h^2 + x^2)} = \sqrt{(0.600^2 + 0.225^2)} = 0.641\text{ m}, \quad \cos\theta = h/R = 0.600/0.641 = 0.936$$

$$R_d = (R_{ME} A / \rho) \cdot (\cos\theta / \pi R^2)$$

$$R_d = [(7.83 \times 10^{-4})(4.91 \times 10^{-4}) / 2700] \times [0.936 / (\pi \times 0.641^2)] = \mathbf{1.033 \times 10^{-10}\ m/s \approx 1.03\ Å/s}$$

Step 3: thickness. The lab manual gives a deposition time of 5–10 min:

$$d = R_d \cdot t: \quad t = 5\text{ min} \rightarrow d \approx \mathbf{31\ nm}; \quad t = 10\text{ min} \rightarrow d \approx 62\text{ nm}$$

Using 1200 °C or 1250 °C instead changes this by less than 1%. Our 5-minute value is a little below the TA's 34.08 nm because our chamber is taller (0.600 m instead of 0.564 m).

Comparison with the gauge. The gauge's 46.4 nm lies inside our 31–62 nm range. It corresponds to t = 46.4 nm ÷ 1.033 Å/s ≈ 450 s ≈ 7.5 min of deposition, which fits the 5–10 min the boat took to empty. As a further check, the evaporated mass in that time, $R_{ME} \cdot A \cdot t \approx 0.17$ g, is about what a few short pieces of Al wire weigh. Reasons the two numbers can differ:

- **T and P_e are estimates.** The boat temperature was not measured. Vapor pressure rises very steeply with temperature, so a temperature error of a few tens of °C changes P_e, and therefore R_d, by a large factor. The rate also changed during the run as the Variac current was raised and the Al ran out.
- **Simplified geometry.** The equation treats the boat as a small flat source and uses only one cosine factor. The source area and the wafer's position in the chamber were estimated.
- **The gauge's own settings.** The quartz crystal monitor works out thickness from mass, using the density (2.70 g/cm^3 for Al), the Z-ratio (1.08 for Al), and a tooling factor. The tooling factor corrects for the crystal sitting closer to or farther from the source than the wafers. If the controller still had another film's settings, its reading would be off by the same factor.
- **Film density.** Evaporated films can be slightly less dense than bulk Al, which makes them a little thicker than the bulk density predicts.

Why the melting point and chamber pressure don't work: P_e must be the vapor pressure of the aluminum itself. Using T = 660 °C and the chamber pressure 10^{-6} Torr gives $R_{ME} \approx 1 \times 10^{-7}$ kg/(m^2·s) and only about 0.004 nm in 5 minutes. That is ten thousand times too thin, which is why the TA updated the parameters.

To measure the true thickness, use a profilometer step height after next week's Al etch.

2. Photoresist used and its thickness

Resist. We used **MEGAPOSIT SPR220** (DuPont, formerly Shipley / Rohm and Haas; sold in the US through Kayaku Advanced Materials) instead of the SJR5740 named in the handout. SPR220 is a thick, positive i-line resist made for 1–10 µm films. The number after the dash gives the nominal thickness: SPR220-3.0 is about 3 µm and SPR220-7.0 is about 7 µm.

Thickness. For spin coating, thickness falls as one over the square root of spin speed ($t \propto 1/\sqrt{\omega}$). SPR220-3.0 gives about 3.0 µm at 2500 rpm, so at 2000 rpm:

$$t \approx 3.0\ \mu\text{m} \times \sqrt{(2500/2000)} \approx 3.3\ \mu\text{m}$$

Cleanroom data report about 3.2 µm at 2000 rpm for SPR220-3.0. We therefore expect **about 3.2–3.4 µm** in the flat centre of the wafer. Soft-baking drives out solvent and shrinks the film by a few percent. (If the bottle was SPR220-7.0, the film would be about 7–8 µm at 2000 rpm. Check the label.)

3. Positive vs. negative photoresist

	Positive resist	Negative resist
Effect of exposure	The exposed areas become **more** soluble in developer (a photoactive compound breaks down, e.g. DNQ to a carboxylic acid in a novolac resin)	The exposed areas become **less** soluble (the polymer crosslinks or hardens)
What stays after developing	The **unexposed** areas, so the pattern is the **same** as the mask's opaque regions	The **exposed** areas, so the pattern is the **reverse** of the mask
Resolution	Better. The film does not swell in developer, so edges stay sharp	Older types swell in solvent developer, which limits resolution. Modern chemically amplified types are much better
Sidewalls and uses	Steep sidewalls. Good for etch masks. The standard choice for IC patterning	Can give undercut (re-entrant) profiles, which are good for lift-off. Can be very thick and tough (SU-8)
Chemical resistance and adhesion	Moderate	Usually stronger. Crosslinked films can be hard to strip

Resists used in industry today, and who makes them:

- **Positive:**
  - **DuPont (formerly Shipley / Dow):** MEGAPOSIT SPR220 (the one we used), S1813 / S1800 series, SPR955.
  - **Merck (EMD Performance Materials, AZ brand):** AZ 1512, AZ 4620, AZ 10XT.
  - Others: TOK (Tokyo Ohka Kogyo) THMR-iP series, and JSR.
- **Negative:**
  - **Kayaku Advanced Materials (formerly MicroChem):** SU-8 and KMPR.
  - **Merck (AZ brand):** AZ nLOF 2020 / 2070, a lift-off resist.
  - Others: Futurrex NR9 series, and micro resist technology ma-N 1400.

4. Is the photoresist the same thickness everywhere?

No. The resist is close to uniform across the flat centre of the wafer, but it varies in two places:

1. **Edge bead (thickest overall).** Resist piles up in a ridge within about 1–3 mm of the wafer rim, and near the flat. Two things cause it. Surface tension holds liquid at the edge as it is flung outward. Solvent also evaporates faster at the rim, so the resist there thickens before it can level out. The bead can be 2–5 times as thick as the film in the centre. That matters next week: it can stop the mask from sitting flat on the wafer during contact exposure.
2. **Over the device's surface relief.** Spun resist tends to level itself out. It is **thicker in low areas** (windows or trenches from earlier steps) and **thinner over raised features**, and thinnest right at the top corners of steps. Exposure and development times must be long enough to clear the thickest parts.

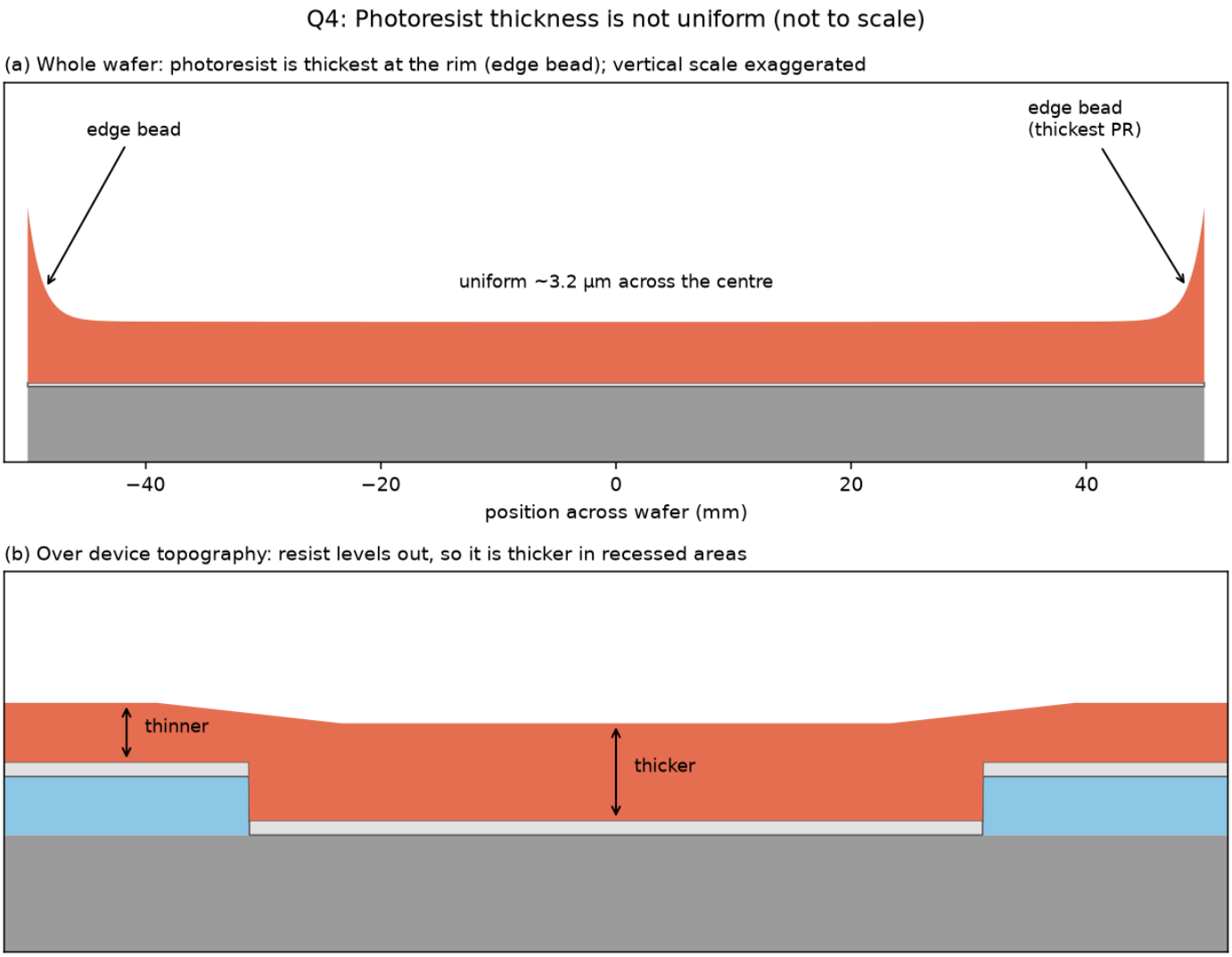


Figure 1. Photoresist thickness across the wafer (edge bead) and over device features. Not to scale.

5. Important cross-sections after this lab

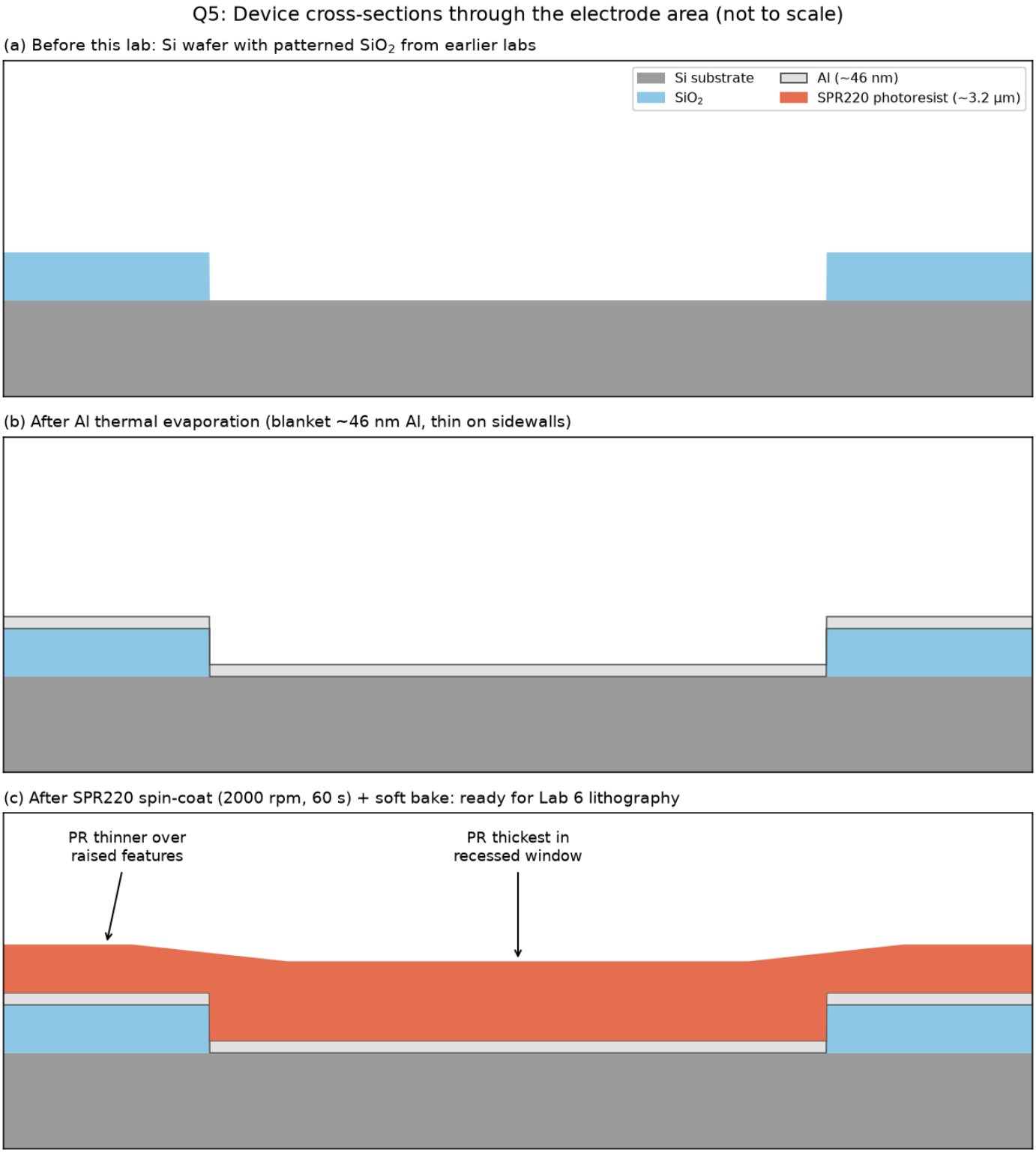


Figure 2. Device cross-sections through the electrode area. Not to scale.

- **(a) Starting point.** The Si wafer with the patterned SiO_2 from earlier labs.
- **(b) After evaporation.** A blanket Al film, about 46 nm thick, covers everything. Evaporation travels in straight lines, so the vertical sidewalls get very little Al. The Al may even break at tall steps.
- **(c) After spin-coating and soft bake.** About 3.2 µm of SPR220 covers the Al. It is thicker in recessed areas and builds up at the wafer edge. The wafer is now ready for next week's lithography, which will

pattern the Al into electrodes.

6. Why pump down to 1×10^{-6} torr before evaporating?

1. **Long mean free path, so atoms fly straight.** The mean free path of gas molecules is $\lambda = kT / (\sqrt{2}\,\pi d^2 P)$. At 10^{-6} torr (1.33×10^{-4} Pa) and room temperature, with $d \approx 0.37$ nm:

   $$\lambda = (1.38 \times 10^{-23} \times 295) / [\sqrt{2} \times \pi \times (3.7 \times 10^{-10})^2 \times 1.33 \times 10^{-4}] \approx 50 \text{ m}$$

   That is about 80 times the 0.6 m path from source to wafer. Only $1 - e^{-0.6/50} \approx 1\%$ of Al atoms hit a gas molecule on the way, so they arrive in a straight line. This gives a predictable, directional film. At higher pressure, the atoms would scatter, lose energy, and form a porous, poorly controlled deposit.
2. **Film purity and adhesion.** Even at 10^{-6} torr, background gas (mostly water vapour and O_2) can cover a surface with one molecular layer in about 1 s. At higher pressures this happens much faster. Aluminum reacts readily with oxygen and water, so a poor vacuum would leave oxide and hydroxide mixed into the film. That raises the resistivity of our electrodes and weakens adhesion. Lower pressure means fewer contaminants arriving compared with Al atoms.
3. **Protecting the hot source.** The tungsten boat and the molten Al run at over 1000 °C. With too much oxygen or water present, the boat and the charge would oxidize, which can burn out the boat and cause spitting.
4. **Stable process and equipment.** A clean, low base pressure makes the deposition rate repeatable. It is also needed for the ion gauge filament and the crystal monitor to work properly.

Sources

- University of Michigan LNF, CEE 200X spinner (SPR220 at 2000 rpm): https://lnf-wiki.eecs.umich.edu/wiki/index.php?title=CEE_200X_photoresist_spinner_1
- UCSB Nanofab, SPR-220-3.0 optimization: https://wiki.nanofab.ucsb.edu/w/images/0/01/SPR-220-3.0_OptimizationNew.pdf
- Kayaku Advanced Materials, MEGAPOSIT SPR220 series: https://kayakuam.com/products/megaposit-spr220-series-thick-i-line-photoresists/
- SPR220 data sheet (UPenn copy): https://www.seas.upenn.edu/~nanosop/documents/SPR_220_Data_Sheet.pdf